Reminders and announcements

- During lectures you can submit questions to: vevox.app ID: 107-018-952
- You can find answers to questions in the COMP1870 Teams Channel (link in Minerva) (along with other discussion).
- Maths Lunch Hours, Mondays and Wednesdays, 1.00pm - 2.00pm. Bragg GR.18.
- My student hour: Tuesdays 1.00pm - 2.00pm, Bragg building foyer.

Revision

- If m and n are positive integers, there are unique integers q and r satisfying $m = qn + r$ and $0 \leq r < n$. We say q is the **quotient** and r is the **remainder** when we divide m by n.
- We write $a \equiv b \pmod{n}$ to mean that n divides $a - b$. This is an equivalence relation.
- A **residue** modulo n is one of the numbers $\{0, 1, 2, \ldots, n-1\}$.
- We write $a \bmod n$ to represent the remainder (residue) we obtain when dividing a by n. Therefore $a \bmod n$ is always in $\{0, 1, 2, \ldots, n-1\}$.

Revision

Repeated squaring modulo n

Question. Calculate 2^{39} mod 9.

i	y_i	z	p
0			
1			
2			
3			
4			
5			

Euclidean algorithm

Let m and n be positive integers. If d divides both m and n, it is a **common divisor**. The **Euclidean algorithm** finds the **greatest common divisor** (written as $\gcd(m, n)$).

- Set r_0 to be m, and r_1 to be n.
- Repeatedly define $r_i = r_{i-2} \bmod r_{i-1}$.
- If $r_t = 0$, then $r_{t-1} = \gcd(m, n)$.

Running time of the Euclidean algorithm

Proposition 9.15. Let m and n be non-negative integers such that $m \geq n$. If $r = m \bmod n$ then $r < m/2$.

Proposition 9.16. Let m and n be non-negative integers satisfying $m \geq n$. Let q be the smallest integer such that $2^q > m$. Define the sequence $r_0, r_1, r_2, \ldots$ by $r_0 = m$, $r_1 = n$, and $r_i = r_{i-2} \bmod r_{i-1}$. Then there is some integer $p \leq 2q$ such that $r_p = 0$.

Euclidean algorithm

Input: Non-negative integers m and n where $m \geq n$.
Output: The greatest common divisor $r_{\text{old}} = \gcd(m, n)$.

$r_{\text{old}} \leftarrow m$
$r_{\text{new}} \leftarrow n$
while $r_{\text{new}} > 0$ **do**
    $t \leftarrow r_{\text{new}}$
    $r_{\text{new}} \leftarrow r_{\text{old}} \bmod t$
    $r_{\text{old}} \leftarrow t$
end while
return r_{old}

The extended Euclidean algorithm

Proposition 9.17. Let m and n be positive integers. There exist integers x and y such that $xm + yn = \gcd(m, n)$.

For paper-and-pen examples, we can find the integers x and y by keeping track of the quotients and remainders as we proceed through the Euclidean algorithm, and then backtrack through the equations, carrying out substitutions.

The extended Euclidean algorithm

Example. Find $\gcd(33, 27)$ and integers x and y such that $\gcd(33, 27) = 33x + 27y$.

The extended Euclidean algorithm

Example. Find $\gcd(62, 23)$ and integers x and y such that $\gcd(62, 23) = 62x + 23y$.

The extended Euclidean algorithm

Input: Non-negative integers m and n where $m \geq n$.
Output: The greatest common divisor $r_{\text{old}} = \gcd(m, n)$ and integers x_{old} and y_{old} such that $\gcd(m, n) = x_{\text{old}}\, m + y_{\text{old}}\, n$.

$r_{\text{old}} \leftarrow m$
$r_{\text{new}} \leftarrow n$
$x_{\text{old}} \leftarrow 1$
$x_{\text{new}} \leftarrow 0$
$y_{\text{old}} \leftarrow 0$
$y_{\text{new}} \leftarrow 1$
while $r_{\text{new}} > 0$ **do**
    $t \leftarrow r_{\text{new}}$
    $(q, r) \leftarrow$ integers such that $r_{\text{old}} = qt + r$ and $0 \leq r < t$
    $r_{\text{new}} \leftarrow r$
    $r_{\text{old}} \leftarrow t$
    $t_x \leftarrow x_{\text{new}}$
    $x_{\text{new}} \leftarrow x_{\text{old}} - qt_x$
    $x_{\text{old}} \leftarrow t_x$
    $t_y \leftarrow y_{\text{new}}$
    $y_{\text{new}} \leftarrow y_{\text{old}} - qt_y$
    $y_{\text{old}} \leftarrow t_y$
end while
return $(r_{\text{old}}, x_{\text{old}}, y_{\text{old}})$

Multiplicative inverses

Let m be a positive integer, and let a and b be residues modulo m.

If $ab \bmod m = 1$, then we say that a and b are **inverses modulo** n.

If a has an inverse modulo m, then we write it as a^{-1}.

Multiplicative inverses

If $\gcd(n, a) = 1$, then we say that n and a are **coprime**

Proposition 9.18. Let n be a positive integer and let a be a residue modulo n. Then a has a multiplicative inverse modulo n if and only if n and a are coprime.

Finding multiplicative inverses

We can use the extended Euclidean algorithm to find the multiplicative inverse of a modulo m.

- Use the extended Euclidean algorithm to find integers x and y such that $xn + ya = 1$.
- Find the residue of y modulo n. This is the number in $0, 1, 2, \ldots, n-1$ that has the same remainder as y modulo n.
- This residue is the inverse of a.

Finding multiplicative inverses

We can use the extended Euclidean algorithm to find multiplicative inverses.

Example. Find the inverse of 8 modulo 11.

Finding multiplicative inverses

Example. Find the inverse of 6 modulo 17.

Pseudorandom number generation

Random numbers are useful in many applications, especially in cryptography.

It is very hard to produce truly random numbers using a computer.

Instead, it is sometimes sufficient to produce a sequence that appears random. Such a sequence is called **pseudorandom**.

Pseudorandom number generation

- Choose n, a very large prime number.
- Choose integers a and c such that $0 < a, c < n$.
- Choose a seed integer x_0 such that $0 \leq x_0 < n$.
- Define x_{i+1} to be

$$x_{i+1} = \begin{cases} (ax_i^{-1} + c) \bmod n & \text{if } x_i \neq 0 \\ c & \text{if } x_i = 0 \end{cases}$$